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Letter

Active Learning FEP Using 3D-QSAR for Prioritizing Bioisosteres in Medicinal Chemistry

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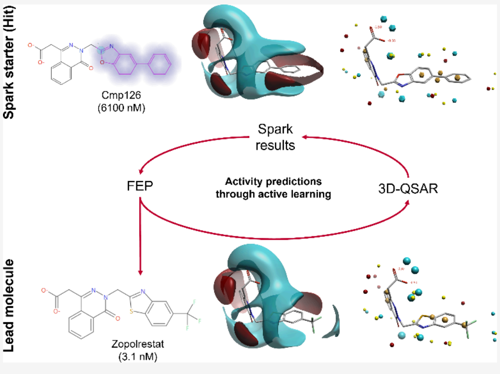
ABSTRACT: Bioisostere replacement is a powerful and popular tool used to optimize the potency and selectivity of candidate molecules in drug discovery. Selecting the right bioisosteres to invest resources in for synthesis and subsequent optimization is key to an efficient drug discovery project. Here we demonstrate how 3D-quantitative structure–activity relationship (3D-QSAR) and relative binding free energy calculations can be combined into an active learning workflow to prioritize molecules from a pool of hundreds of bioisosteres. We demonstrate on a human aldose reductase test case that the use of this workflow can rapidly locate the strongest-binding bioisosteric replacements with a relatively modest computational cost.



KEYWORDS: *active learning, bioisostere, Spark, 3D-QSAR, FEP*

Hit-to-lead optimization requires a delicate balancing of potency, safety, physicochemical properties and intellectual property to transform hits into potent lead compounds. Bioisostere replacement is a well-established and commonly applied method to generate lead-like compounds from a starting molecule.[1] The Spark[2] bioisostere and R-group replacement software uses Cresset's electrostatic and shape similarity method[3] to generate hundreds of viable replacements for part of a hit molecule, but the onus lies on the user to select a handful of these to focus the resources on to increase the success rate of a medicinal chemistry/drug discovery project. Although the Spark scoring algorithms perform well at suggesting replacements which will preserve some degree of bioactivity, such docking and ligand similarity methods are generally not capable of providing accurate predictions of the exact level of activity.[4−6]

A number of approaches are available for more precise rank ordering of molecules by protein–ligand binding affinity. The gold standard is rigorous physics-based methods such as free energy perturbation[7,8] (FEP). However, these can be extremely computationally intensive, especially when considering data set of several hundreds of molecules. More heuristic methods such as quantitative structure–activity relationship (QSAR) modeling allow for faster predictions but require a significant amount of training data and hence are not generally applicable in early hit-to-lead stages. Here, we demonstrate that these two methods can be linked into a cyclic workflow to get the best of both the worlds to improve their predictive power and use them to triage the final set of lead molecules. This workflow, often referred to as active learning, involves using machine learning models to steer the search of identifying the top ranked molecules while significantly reducing the number of FEP calculations performed to overcome time and cost related challenges.[9,10]

There have been a few publications evaluating and applying active learning to FEP using different design strategies, data set sizes, sources of activity and number of iterations of active learning.[10−13] These have invariably used traditional 2D-QSAR using molecular fingerprint descriptors. Given that the Spark algorithm focuses on identifying bioisosteric replacements with high electrostatic and shape similarity to the target molecule in 3D, all of the molecules are well-aligned within a given protein context and have a high degree of 3D similarity. The resulting data set is therefore ideally suited to 3D-QSAR. In this paper we show that the additional information content from utilizing 3D descriptors allows the QSAR method to be more predictive with a small amount of training data, allowing active learning to proceed with a relatively modest number of activity evaluations (Figure 1).

Diabetes is a chronic metabolic disease and has a significant global impact to the extent of having a globally agreed target to arrest the rise of diabetes and obesity by 2025 (https://www.

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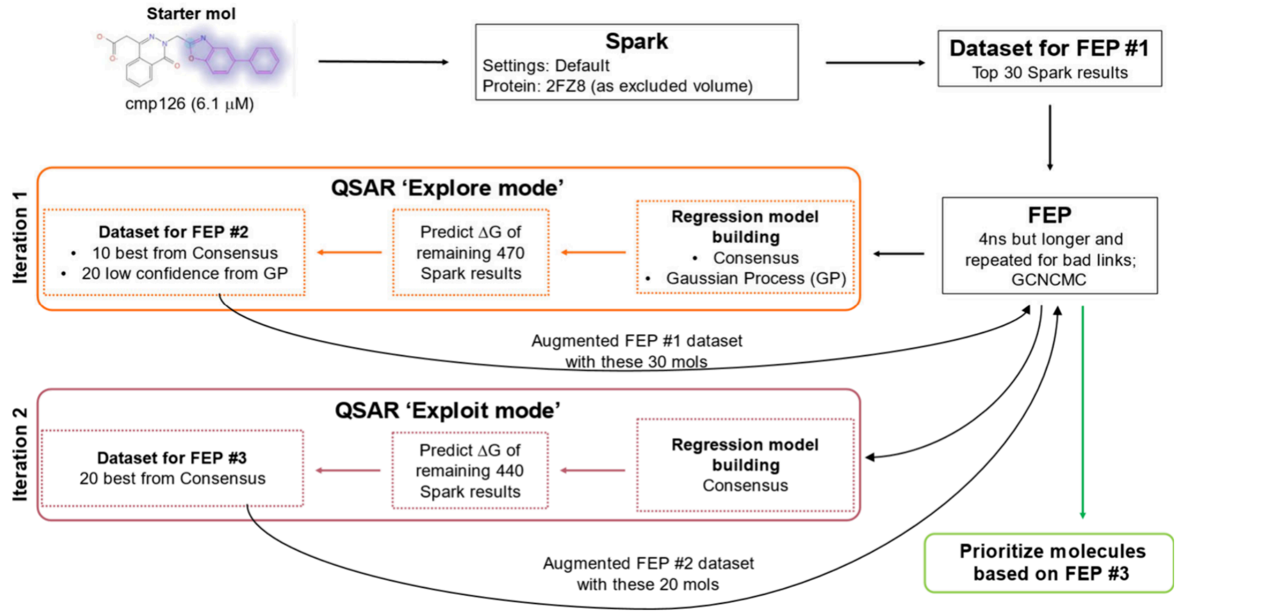

Figure 1. Overview of our active learning workflow showing the different stages from hit to lead identification.

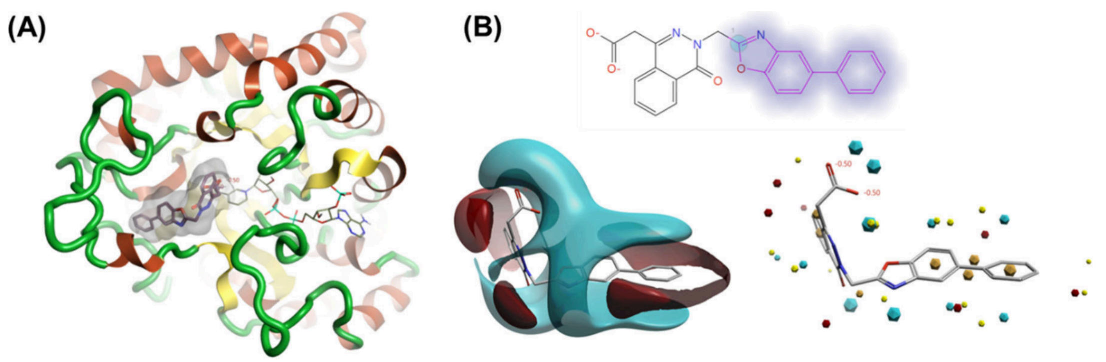

Figure 2. (A) The ribbon representation of human aldose reductase X-ray structure (PDB: 2FZ8) with crystallographic bound Zopolrestat (molecular surface only view) and the starter molecule (represented in panel B) to Spark embedded in it (sticks). The bound cofactor nicotinamide adenine dinucleotide phosphate is also shown next to it. (B) 2D representation of the starter molecule (cmp126) with a pIC50 of 5.21 (6.1 μM) against human ALR2. The region selected for R-group replacement in Spark is highlighted by halos. 3D molecular electrostatic potential (MEP) and field points[3] of the molecule are shown in the bottom left and bottom right, respectively. Color code for MEP and field points: red for positive, blue for negative, yellow for shape, and orange for hydrophobic.

who.int/health-topics/diabetes). The enzyme aldose reductase (ALR2) reduces glucose to sorbitol in the presence of NADPH, which is the first and rate-determining step in polyol pathway. In a hyperglycaemic state, the overactivity of ALR2 leads to an accumulation of intracellular sorbitol, depletion of cytoplasmic NADPH, oxidative stress culminating in the emergence and progression of diabetic complications such as nephropathy, neuropathy, retinopathy, atherosclerosis and others cardiovascular disorders.[14,15] ALR2 inhibitors are considered a promising tool to treat such diabetic complications and the extensive structure−activity relationship data[16−19] available on ALR2 inhibitors makes it an ideal subject to evaluate our workflow.

Imagining a hypothetical hit-to-lead stage in a typical drug discovery project, we picked cmp126, a low potency binder (pIC50 of 5.21) from the Pfizer substituted thiazole series[17] as a starter molecule for our workflow. We used cmp126 as the starter for R-group replacement experiment (see Methods) using Spark to generate a pool of bioisosteres of the phenyl benzoxazole (Figure 2).

Spark scores bioisosteres by evaluating the shape and electrostatic similarity of the bioisosterically replaced molecule against the starting molecule. The similarity assessment takes place in "product space" rather than "fragment space" as the electrostatic properties of a fragment can vary markedly depending on the molecular context. In this case, Spark identified a number of replacements for the phenyl benzoxazole which were known from the literature, including benzoxazole, benzothiazole and benzodiazole substituted molecules.[17,18] A total of 32 ALR2 inhibitors with reported activity data (pIC50 spanning 5.21 to 9) were present in the pool of 500 suggested bioisosteric replacements (Figure S1). While Spark was very successful at identifying bioisosteric replacements that retained a degree of bioactivity, the similarity-based scoring method is not accurate enough to correctly rank the suggested replacements (Figure S2). Relative binding free energy methods could provide a significantly more accurate ranking, but applying these to all 500 bioisosteric suggestions would incur a significant computational cost.

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The initial stage of the active learning protocol requires the generation of predicted activity data for an initial cohort of molecules to train a machine learning model. Literature protocols often recommend that this initial cohort be in the order of 100 molecules, but the use of more information-rich 3D descriptors should allow this to be reduced. Accordingly, the initial cohort was the top 30 from the Spark results ordered by similarity score. The standard Flare FEP protocol[20,21] (see Methods) was followed with cmp126 marked as the reference molecule with known experimental activity.

FEP predicted binding free energies (ΔG) of these 30 molecules were then used to build two 3D-QSAR regression models (see Methods): Consensus and Gaussian Process. Both the models showed good predictive capabilities as demonstrated by their evaluation metrics on the cross-validation training sets. The Q^2 values were around 0.89 and 0.82 for Consensus and Gaussian Process models, respectively, while their respective Kendall's tau was around 0.79 and 0.74 (Figure S3(A)). Given that 2D QSAR methods are more commonly used, the question arises of whether a 2D-descriptor based model would perform as well or better in this context. Based on a benchmark test using this same training set, we discovered that these 3D-QSAR models outperformed the 2D-QSAR models built with standard RDKit descriptors. Combining 2D with 3D descriptors showed performance, at best, similar to using only 3D descriptors. This finding confirms our initial hypothesis that incorporating additional information from 3D descriptors enhances the predictive power of QSAR, even with a limited amount of training data (Tables S1 and S2). Both these 3D-QSAR models were applied to the remaining 470 molecules in the data set. For the early rounds of active learning, it is suggested that optimal results are obtained by using the ML predictions to maximize the information content of the data set (the so-called "explore" mode) rather than just taking the highest-predicted molecules. Accordingly, we picked the top 10 ligands based on the Consensus regression model, alongside 20 molecules showing the largest standard deviations in the Gaussian Process regression model. The intention was to provide a balance of high-predicted-activity and high-uncertainty molecules for the following active learning round. This cohort was added to the previous FEP network and their predicted binding free energies were then determined by relative binding free energy calculations as before, and a new set of 3D-QSAR models built.

The consensus 3D-QSAR model from this round retained high predictivity (Q^2 and Kendall's tau of around 0.8 and 0.7, respectively. Figure S3(B)) but was expected to have a wider domain of applicability than the first model due to its construction. While it would be possible to do multiple rounds of active learning in "explore" mode, our intention was to assess how well the active learning workflow would work assuming very limited computational resources, and as a result a single final active learning round was performed.

In this final round, the 3D-QSAR model was applied to the remaining 440 molecules in the Spark result data set and the top 20 molecules were taken into free energy calculations. At the end of the active learning process, therefore, a total of 80 molecules had been assessed using free energy calculations, representing only 16% of the pool of bioisosteric replacements identified by Spark.

Examining the results, the active learning protocol had done extremely well at building a predictive model and identifying activity-improving bioisostere replacements for the starting molecule. Comparing the model predictions to the computed binding free energies demonstrated that the model had been extremely predictive and effective in identifying tight-binding bioisosteric replacements (Figure 3).

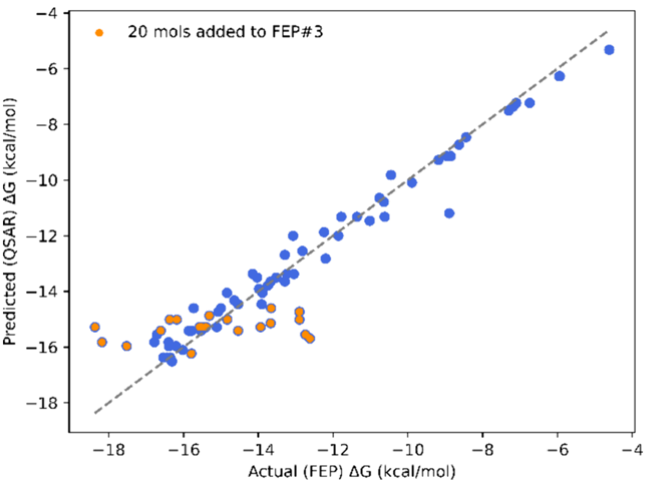

Figure 3. Scatter plot highlighting the high predictive power of the final 3D-QSAR model. The top 20 molecules predicted using the model taken forward to FEP #3 are highlighted in orange, while molecules used to train it are shown in blue.

Around 45% of the known actives in the Spark data set had been chosen for free energy calculations (an enrichment factor of 2.8). Moreover, the final ranking from the active learning workflow retrieved 10 known actives in the top 20, including Zopolrestat (which entered the clinic)[18] in position 3, while the original Spark ranking retrieved only 4 (Figure 4, also see Tables S3 and S4).

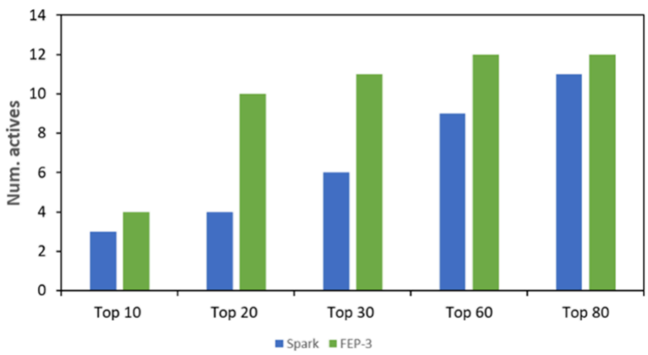

Figure 4. Histogram plot of the number of all actives among the top-ranked molecules retrieved from Spark and after the last stage of our active learning workflow. See Figure S4 for a breakdown of this at different stages of the workflow.

The area under ROC curves based on the top 80 results from Spark and our active learning workflow highlight this improved performance of our method (ROC AUC of 0.88) compared to directly picking actives from the bioisostere replacement experiment (ROC AUC of 0.64) (Figure 5).

The top 20 compounds prioritized from our active learning FEP workflow had several benzothiazole and benzoxazole side chain containing molecules, a series previously tested by Mylari et al.[17,18] and found to yield actives (Figure 6). On analyzing the binding mode of the top 5 molecules in their equilibrated

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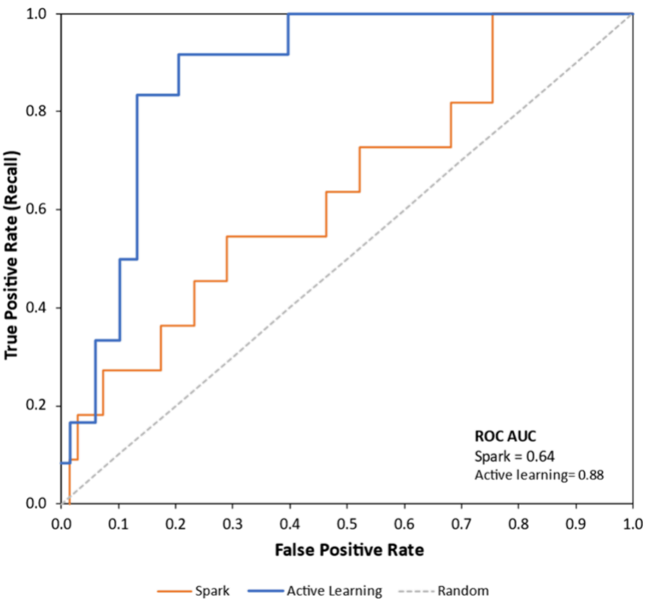

Figure 5. Area under the ROC curve considering the top 80 results highlighting the early enrichment of actives in our active learning workflow (blue) in contrast to using none (orange) in picking actives from the Spark pool of results. See Figure S5 for the ROC AUC on the complete data set. Molecules with reported pIC50 > 6 are considered actives.

complexes with ALR2, we found all of them making (i) perfect π-stacking interactions with their bioisosteric replacement and Trp111 in the hydrophobic specificity pocket and (ii) short contacts (hydrogen bonding) between their carboxylic acid headgroup and side chains of Tyr48, His110, Trp111 in the catalytic anion binding site, a binding mode highly selective to aldose reductase and not aldehyde reductase[22−24] (Figure 7). Also, the high level of plasticity of the specificity pocket owing to its residues Phe122, Ala299 and Leu300 being able to populate different rotameric states and thereby opening variably to accommodate different inhibitors opens up the avenue to explore the bioisosteres suggested by Spark that were ranked high but for which there is no reported activity data.[24−26]

Overall, our active learning workflow demonstrates how the synergistic application of the fast predictive power of 3D-QSAR and the rigorous affinity prediction and ranking of Flare FEP can significantly boost the success rate of selecting potential leads (actives) from a pool of hundreds of bioisosteres while making use of only a fraction of computational cost that would have otherwise been required (this workflow used less than 20% of the compute time needed to generate FEP results for all 500 Spark results). In this case of prioritizing ALR2 inhibitors from a pool of 500 lead-like compounds suggested by Cresset Spark, our workflow increased the success rate by nearly 2.5 times. The number of cycles of active learning was found to be minimal to obtain a reasonable improvement without eating too much into computational resources. We believe our active learning workflow has a much more widespread application and can be applied to other systems amenable to 3D-QSAR and FEP.

■ METHODS

Bioisostere Replacement Experiment. The ChEMBL database[27] was searched for human ALR2 inhibitors with activity data (IC50) from the same research group. Cmp126 (pIC50 5.21) was chosen as the hit in our hit-to-lead optimization workflow. The X-ray

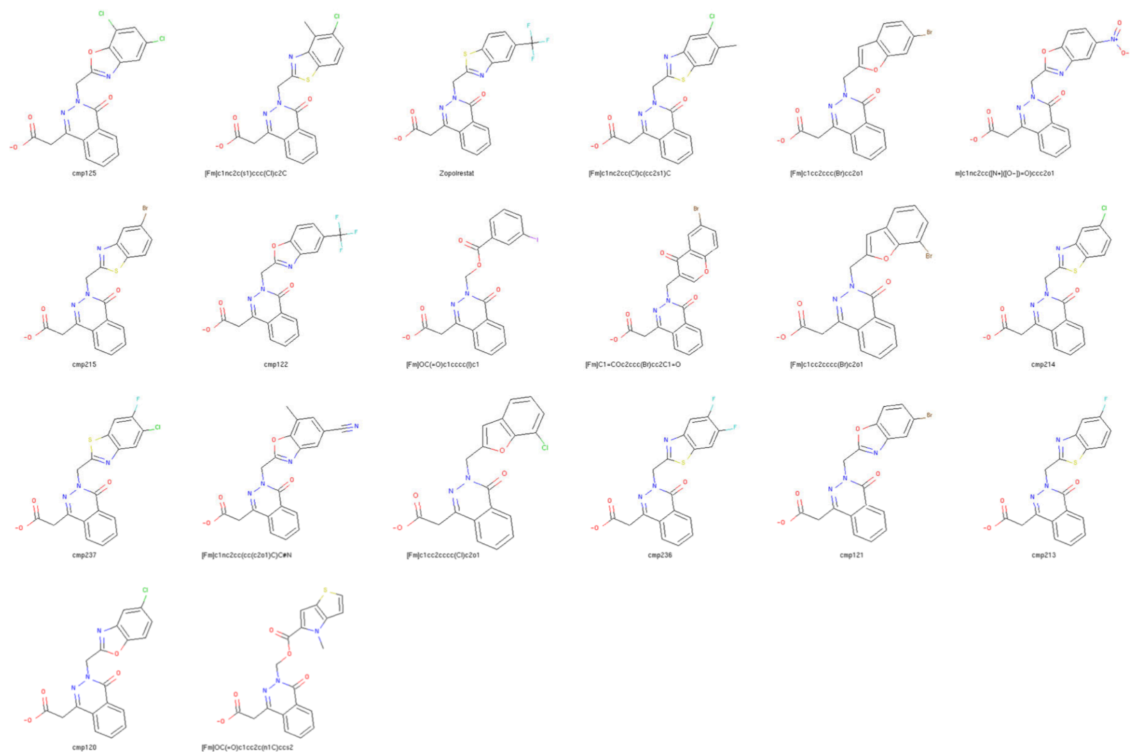

Figure 6. Top 20 molecules from our active learning workflow rank ordered as per their FEP predicted binding affinities from top left to bottom right. Molecules are labeled according to their compound names for known actives,[17,18] while novel molecules obtained from Spark with no reported activity start with the prefix [Fm].

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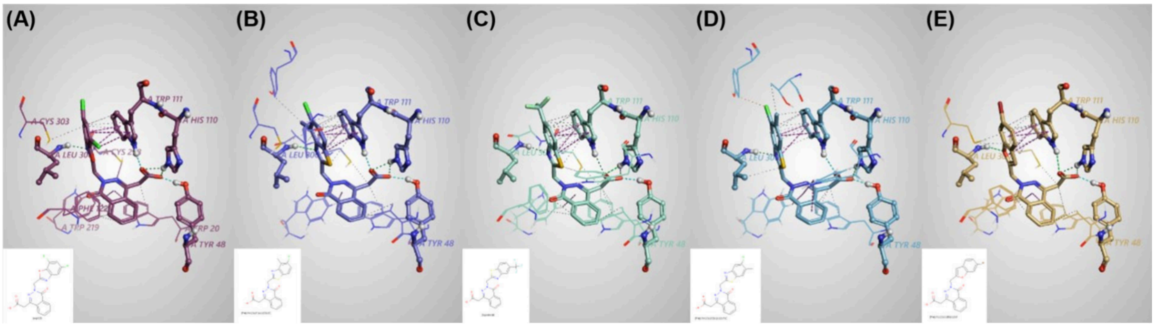

Figure 7. Binding site interactions of the top 5 molecules from our active learning workflow with ALR2 protein in their equilibrated complexes from Flare FEP. All complexes show a perfect π-stacking interaction made by the bioisosteric replacements with TRP111 and the short contacts by the carboxylic acid headgroup in the catalytic anion binding site. The molecules are ordered as per their rank from left to right. (A) and (C) are known ALR2 inhibitors cmp125 and Zopolrestat, respectively.[17]

structure of human ALR2 complexed with Zopolrestat (PDB: 2FZ8)[28] was imported into Flare[29] V8.0, prepared and used as the starting point. Cmp126 was aligned to the cocrystallized Zopolrestat based on substructure alignment within Flare. The aligned compound (cmp126) was then used as the starter molecule for R-group replacement using Cresset Spark.[2] The prepared protein with all crystallographic waters removed was imported into Spark with the input protonation retained and used as an excluded volume to guide the search. The phenyl benzoxazole in the starter molecule was selected as the region to replace (Figure 2(B)). The ChEMBL_common and Commercial Common and Very Common databases were searched with default settings. These Spark databases of fragments contained 95,568 fragments for R-group replacement experiments derived from compounds reported in literature (ChEMBL database) (https://www.ebi.ac.uk/chembl/) and commercial compounds (VeryCommon and Common) (https://www.emolecules.com/).

The R-group replacement experiment using the Spark settings detailed above generated 500 suggestions based on 50% field and 50% shape similarity, with Bioisostere Factor (BIF%) values ranging from 56 to 82. The BIF% is a scaled score such that the original molecule scores 100% and the score for deleting the part to be replaced is zero. Positive values indicate favorable bioisosteres, negative values reflect a replacement that can reproduce the geometry of the original molecule but is a poor mimic of the deleted moiety.

Free Energy Perturbation Calculations. Free energy calculations were carried out with the standard protocol using Flare V8.0. The crystal structure of human ALR2 (PDB: 2FZ8) was prepared using Flare and protonation states were assigned using the TSAR algorithm.[30] The protonation states of the ligands were retained as obtained from Spark without any postprocessing. The AMBER FF14SB[31] force field was used to describe the proteins and OpenFF 2.1.0[32] (https://zenodo.org/records/7889050) for the ligand, with AM1-BCC charges. Chiral, rotamer and tautomeric states were checked and included in the FEP map where needed. The systems were solvated using the TIP3P water model with a truncated octahedron solvation box with a solvent box buffer of 12 Å. GCNCMC[33] sampling protocol with a 4 Å buffer size for the sampling sphere was used during equilibration phase only. Na^+ and Cl^- ions were added as required to neutralize the system. Bonds involving hydrogen atoms were constrained using the SHAKE algorithm, hydrogen mass repartitioning was done, and a time step of 4 fs was used throughout all simulations. Long-range electrostatic interactions were treated using the particle-mesh Ewald (PME) with a nonbonded cutoff of 9 Å. Systems were equilibrated using OpenMM package[34] at $\lambda = 0$ following the default protocol.

The perturbation networks were created using a customized version of LOMAP[35] implemented in Flare and intermediates were added automatically where possible when the link score is below a threshold of 0.4. Also, the maximum distance from any ligand to the known active (cmp126) was set to 3 in the production mode. Each perturbation was carried out in both forward and reverse directions as two independent simulations. The number of lambda windows for each transformation was automatically assessed using the adaptive lambda window scheduling.[36] Each window was subjected to 4 ns MD simulation though additional λ windows and longer simulations were used for more challenging perturbations characterized by large hysteresis. Failed links or links with large hysteresis were repeated. Electrostatic and van der Waals interactions were described using soft-core potentials. The input files describing the perturbation were generated using BioSimSpace[37] and simulations were performed using SOMD.[21] Free energies and their errors were calculated using MBAR.[38] The final change in the free energy of binding for each ligand pair was calculated by averaging results obtained for the perturbation carried out in both directions. ΔG values were computed from the network according to the method of Xu.[39]

The performance of Flare FEP on ALR2 data set was assessed using a set of 10 compounds randomly chosen from among those with known activities.[17,18] Our protocol achieved a coefficient of determination (R^2) of 0.8 and a mean unsigned error (MUE) of 1.6 kcal/mol demonstrating its suitability and reliability for this case study (Figure S6). We also evaluated the effect of using an alternate receptor structure in FEP and found the FEP data, in this case, to be relatively independent of the receptor structure (see Supporting Information section "Effect of receptor structure in FEP" and Figure S7). As part of the active learning workflow, FEP was performed on a total of 109 ligands including intermediates (249 links, 499 perturbations). Ligands selected at each stage of the workflow for evaluation with FEP were added to the FEP network generated in the previous round, thereby reusing the generated networks and calculated links and saving on computational resources.

Quantitative Structure−Activity Relations Model Building. Consensus and Gaussian Process regression models were built based on the FEP-derived binding free energy estimates along with Cresset 3D shape and electrostatic descriptors.[3] The consensus regression model was made of SVM Regression, Random Forest Regression, MLP Regression, and Gaussian Process Regression. The predictive performance of the models was assessed using k-fold cross-validation with k = 5.

To generate the 3D QSAR descriptors based on the aligned data set, field points were computed on all molecules.[3] The electrostatic field points of all training set molecules were combined and clustered by taking the largest field point and removing all other field points within 1 Å of it, repeating for the second-largest remaining field point and so forth. The final set of field points comprise a point cloud around the training set. For each molecule in the training set, the electrostatic potential was then computed at each of the positions in the point cloud, to create an electrostatic descriptor vector for that molecule. This process was then repeated, but instead of using field point positions the positions of heavy atoms were used, and the descriptor value was set to 1 if the position was inside a molecule and 0 if it was outside, to produce a binary steric descriptor vector for each molecule. The electrostatic and steric descriptor matrices were block-scaled and then used for the machine learning process.[40]

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Safety Statement. This study was conducted using computational methods only and none of the data generated involve any laboratory experiments. Therefore, there are no associated risks or safety concerns related to handling chemicals, reactions or equipment.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsmedchemlett.4c00554.

Additional experiments, analysis graphs, and 2D representations of known molecules in the pool of bioisosteric replacements (PDF)

Structures of all 500 bioisosteric replacements, along with their FEP and QSAR predicted activity values, where calculated (ZIP)

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Author Contributions

The manuscript was written through contributions of all authors, who have also given approval to the final version of the manuscript.

Notes

The authors declare the following competing financial interest(s): V.K.R. and M.H. are employees of Cresset. M.D.M. is a board member and shareholder of Cresset.

■ ABBREVIATIONS

ALR	Aldose Reductase
AUC	Area Under Curve
BIF	Bioisostere Factor
FEP	Free Energy Perturbation
GCNCMC	Grand Canonical Nonequilibrium Candidate Monte Carlo
GPU	Graphics Processing Unit
LOMAP	Lead Optimization Mapper
MBAR	Multistate Bennett Acceptance Ratio
ML	Machine Learning
MLP	Multilayer Perceptron
MUE	Mean Unsigned Error
NADPH	Nicotinamide Adenine Dinucleotide Phosphate
OpenFF	Open Force Field
PDB	Protein Data Bank
PME	Particle Mesh Ewald
QSAR	Quantitative Structure–Activity Relationship
ROC	Receiver Operating Characteristic
SOMD	Sparse Organized Molecular Dynamics
SVM	Support Vector Machine

■ REFERENCES

(1) Meanwell, N. A. The Influence of Bioisosteres in Drug Design: Tactical Applications to Address Developability Problems. In *Tactics in Contemporary Drug Design*; Meanwell, N. A., Ed.; Topics in Medicinal Chemistry, Vol. *9*; Springer, Berlin, Heidelberg, 2013; pp 283–381. DOI: 10.1007/7355_2013_29.

(2) Spark. https://www.cresset-group.com/software/spark/.

(3) Cheeseright, T.; Mackey, M.; Rose, S.; Vinter, A. Molecular Field Extrema as Descriptors of Biological Activity: Definition and Validation. *J. Chem. Inf. Model.* **2006**, *46* (2), 665–676.

(4) Warren, G. L.; Andrews, C. W.; Capelli, A.-M.; Clarke, B.; LaLonde, J.; Lambert, M. H.; Lindvall, M.; Nevins, N.; Semus, S. F.; Senger, S.; Tedesco, G.; Wall, I. D.; Woolven, J. M.; Peishoff, C. E.; Head, M. S. A Critical Assessment of Docking Programs and Scoring Functions. *J. Med. Chem.* **2006**, *49* (20), 5912–5931.

(5) Pantsar, T.; Poso, A. Binding Affinity via Docking: Fact and Fiction. *Molecules* **2018**, *23* (8), 1899.

(6) Breznik, M.; Ge, Y.; Bluck, J. P.; Briem, H.; Hahn, D. F.; Christ, C. D.; Mortier, J.; Mobley, D. L.; Meier, K. Prioritizing Small Sets of Molecules for Synthesis through In-Silico Tools: A Comparison of Common Ranking Methods. *ChemMedChem.* **2023**, *18* (1), No. e202200425.

(7) Wang, L.; Wu, Y.; Deng, Y.; Kim, B.; Pierce, L.; Krilov, G.; Lupyan, D.; Robinson, S.; Dahlgren, M. K.; Greenwood, J.; Romero, D. L.; Masse, C.; Knight, J. L.; Steinbrecher, T.; Beuming, T.; Damm, W.; Harder, E.; Sherman, W.; Brewer, M.; Wester, R.; Murcko, M.; Frye, L.; Farid, R.; Lin, T.; Mobley, D. L.; Jorgensen, W. L.; Berne, B. J.; Friesner, R. A.; Abel, R. Accurate and Reliable Prediction of Relative Ligand Binding Potency in Prospective Drug Discovery by Way of a Modern Free-Energy Calculation Protocol and Force Field. *J. Am. Chem. Soc.* **2015**, *137* (7), 2695–2703.

(8) Mey, A. S. J. S.; Allen, B. K.; Bruce Macdonald, H. E.; Chodera, J. D.; Hahn, D. F.; Kuhn, M.; Michel, J.; Mobley, D. L.; Naden, L. N.; Prasad, S.; Rizzi, A.; Scheen, J.; Shirts, M. R.; Tresadern, G.; Xu, H. Best Practices for Alchemical Free Energy Calculations [Article v1.0]. *Living J. Comput. Mol. Sci.* **2020**, *2* (1), 18378.

(9) Reker, D.; Schneider, G. Active-Learning Strategies in Computer-Assisted Drug Discovery. *Drug Discovery Today* **2015**, *20* (4), 458–465.

(10) Thompson, J.; Walters, W. P.; Feng, J. A.; Pabon, N. A.; Xu, H.; Maser, M.; Goldman, B. B.; Moustakas, D.; Schmidt, M.; York, F. Optimizing Active Learning for Free Energy Calculations. *Artif. Intell. Life Sci.* **2022**, *2*, No. 100050.

(11) Gusev, F.; Gutkin, E.; Kurnikova, M. G.; Isayev, O. Active Learning Guided Drug Design Lead Optimization Based on Relative Binding Free Energy Modeling. *ChemRxiv* **2022**, 1 DOI: 10.26434/chemrxiv-2022-krs1t.

(12) Khalak, Y.; Tresadern, G.; Hahn, D. F.; de Groot, B. L.; Gapsys, V. Chemical Space Exploration with Active Learning and Alchemical Free Energies. *J. Chem. Theory Comput.* **2022**, *18* (10), 6259–6270.

(13) Burger, P. B.; Hu, X.; Balabin, I.; Muller, M.; Stanley, M.; Joubert, F.; Kaiser, T. M. FEP Augmentation as a Means to Solve Data Paucity Problems for Machine Learning in Chemical Biology. *J. Chem. Inf. Model.* **2024**, *64*, 3812.

(14) Thakur, S.; Gupta, S. K.; Ali, V.; Singh, P.; Verma, M. Aldose Reductase: A Cause and a Potential Target for the Treatment of Diabetic Complications. *Arch. Pharm. Res.* **2021**, *44* (7), 655–667.

(15) Jannapureddy, S.; Sharma, M.; Yepuri, G.; Schmidt, A. M.; Ramasamy, R. Aldose Reductase: An Emerging Target for Development of Interventions for Diabetic Cardiovascular Complications. *Front. Endocrinol.* **2021**, *12*, No. 636267.

(16) Mylari, B. L.; Armento, S. J.; Beebe, D. A.; Conn, E. L.; Coutcher, J. B.; Dina, M. S.; O'Gorman, M. T.; Linhares, M. C.; Martin, W. H.; Oates, P. J.; Tess, D. A.; Withbroe, G. J.; Zembrowski, W. J. A Novel Series of Non-Carboxylic Acid, Non-Hydantoin Inhibitors of Aldose Reductase with Potent Oral Activity in Diabetic Rat Models: 6-(5-Chloro-3-Methylbenzofuran-2-Sulfonyl)-2 *H* -Pyridazin-3-One and Congeners. *J. Med. Chem.* **2005**, *48* (20), 6326–6339.

(17) Mylari, B. L.; Beyer, T. A.; Scott, P. J.; Aldinger, C. E.; Dee, M. F.; Siegel, T. W.; Zembrowski, W. J. Potent, Orally Active Aldose Reductase Inhibitors Related to Zopolrestat: Surrogates for Benzothiazole Side Chain. *J. Med. Chem.* **1992**, *35* (3), 457−465.

(18) Mylari, B. L.; Larson, E. R.; Beyer, T. A.; Zembrowski, W. J.; Aldinger, C. E.; Dee, M. F.; Siegel, T. W.; Singleton, D. H. Novel, Potent Aldose Reductase Inhibitors: 3,4-Dihydro-4-Oxo-3-[[5-(Trifluoromethyl)-2-Benzothiazolyl]Methyl]-1-Phthalazineacetic Acid (Zopolrestat) and Congeners. *J. Med. Chem.* **1991**, *34* (1), 108−122.

(19) Mylari, B. L.; Armento, S. J.; Beebe, D. A.; Conn, E. L.; Coutcher, J. B.; Dina, M. S.; O'Gorman, M. T.; Linhares, M. C.; Martin, W. H.; Oates, P. J.; Tess, D. A.; Withbroe, G. J.; Zembrowski, W. J. A Highly Selective, Non-Hydantoin, Non-Carboxylic Acid Inhibitor of Aldose Reductase with Potent Oral Activity in Diabetic Rat Models: 6-(5-Chloro-3-Methylbenzofuran- 2-Sulfonyl)-2- *H* -Pyridazin-3-One. *J. Med. Chem.* **2003**, *46* (12), 2283−2286.

(20) Flare FEP. https://www.cresset-group.com/software/flare-fep/.

(21) Woods, C. J.; Hedges, L. O.; Mulholland, A. J.; Malaisree, M.; Tosco, P.; Loeffler, H. H.; Suruzhon, M.; Burman, M.; Bariami, S.; Bosisio, S.; Calabro, G.; Clark, F.; Mey, A. S. J. S.; Michel, J. Sire: An Interoperability Engine for Prototyping Algorithms and Exchanging Information between Molecular Simulation Programs. *J. Chem. Phys.* **2024**, *160* (20), 202503 DOI: 10.1063/5.0200458.

(22) El-Kabbani, O.; Wilson, D. K.; Petrash, J. M.; Quiocho, F. A. Structural Features of the Aldose Reductase and Aldehyde Reductase Inhibitor-Binding Sites. *Mol. Vis.* **1998**, *4* (4), 19.

(23) Rees-Milton, K. J.; Jia, Z.; Green, N. C.; Bhatia, M.; El-Kabbani, O.; Flynn, T. G. Aldehyde Reductase: The Role of C-Terminal Residues in Defining Substrate and Cofactor Specificities. *Arch. Biochem. Biophys.* **1998**, *355* (2), 137−144.

(24) Urzhumtsev, A.; Tête-Favier, F.; Mitschler, A.; Barbanton, J.; Barth, P.; Urzhumtseva, L.; Biellmann, J.-F.; Podjarny, A.; Moras, D. A 'Specificity' Pocket Inferred from the Crystal Structures of the Complexes of Aldose Reductase with the Pharmaceutically Important Inhibitors Tolrestat and Sorbinil. *Structure* **1997**, *5* (5), 601−612.

(25) Sotriffer, C. A.; Krämer, O.; Klebe, G. Probing Flexibility and "Induced-fit" Phenomena in Aldose Reductase by Comparative Crystal Structure Analysis and Molecular Dynamics Simulations. *Proteins Struct. Funct. Bioinforma.* **2004**, *56* (1), 52−66.

(26) Joachimiak, A.; Podjarny, A.; Cachau, R. E.; Schneider, T.; Van Zandt, M. Subatomic and Atomic Crystallographic Studies of Aldose Reductase: Implications for Inhibitor Binding. *Cell. Mol. Life Sci. CMLS* **2004**, *61* (7−8), 763−773.

(27) Zdrazil, B.; Felix, E.; Hunter, F.; Manners, E. J.; Blackshaw, J.; Corbett, S.; de Veij, M.; Ioannidis, H.; Lopez, D. M.; Mosquera, J. F.; Magarinos, M. P.; Bosc, N.; Arcila, R.; Kizilören, T.; Gaulton, A.; Bento, A. P.; Adasme, M. F.; Monecke, P.; Landrum, G. A.; Leach, A. R. The ChEMBL Database in 2023: A Drug Discovery Platform Spanning Multiple Bioactivity Data Types and Time Periods. *Nucleic Acids Res.* **2024**, *52* (D1), D1180−D1192.

(28) Steuber, H.; Zentgraf, M.; Gerlach, C.; Sotriffer, C. A.; Heine, A.; Klebe, G. Expect the Unexpected or Caveat for Drug Designers: Multiple Structure Determinations Using Aldose Reductase Crystals Treated under Varying Soaking and Co-Crystallisation Conditions. *J. Mol. Biol.* **2006**, *363* (1), 174−187.

(29) Flare. https://www.cresset-group.com/software/flare/ (accessed 2024-01-03).

(30) Stroganov, O. V.; Novikov, F. N.; Zeifman, A. A.; Stroylov, V. S.; Chilov, G. G. TSAR a New Graph Theoretical Approach to Computational Modeling of Protein Side Chain Flexibility Modeling of Ionization Properties of Proteins. *Proteins Struct. Funct. Bioinforma.* **2011**, *79* (9), 2693−2710.

(31) Maier, J. A.; Martinez, C.; Kasavajhala, K.; Wickstrom, L.; Hauser, K. E.; Simmerling, C. ff14SB: Improving the Accuracy of Protein Side Chain and Backbone Parameters from ff99SB. *J. Chem. Theory Comput.* **2015**, *11* (8), 3696−3713.

(32) Boothroyd, S.; Behara, P. K.; Madin, O.; Hahn, D.; Jang, H.; Gapsys, V.; Wagner, J.; Horton, J.; Dotson, D.; Thompson, M.; Maat, J.; Gokey, T.; Wang, L.-P.; Cole, D.; Gilson, M.; Chodera, J.; Bayly, C.; Shirts, M.; Mobley, D. Development and Benchmarking of Open Force Field 2.0.0 — the Sage Small Molecule Force Field. *ChemRxiv* **2022**, 1 DOI: 10.26434/chemrxiv-2022-n2z1c.

(33) Melling, O. J.; Samways, M. L.; Ge, Y.; Mobley, D. L.; Essex, J. W. Enhanced Grand Canonical Sampling of Occluded Water Sites Using Nonequilibrium Candidate Monte Carlo. *J. Chem. Theory Comput.* **2023**, *19* (3), 1050−1062.

(34) Eastman, P.; Swails, J.; Chodera, J. D.; McGibbon, R. T.; Zhao, Y.; Beauchamp, K. A.; Wang, L.-P.; Simmonett, A. C.; Harrigan, M. P.; Stern, C. D.; Wiewiora, R. P.; Brooks, B. R.; Pande, V. S. OpenMM 7: Rapid Development of High Performance Algorithms for Molecular Dynamics. *PLoS Comput. Biol.* **2017**, *13* (7), No. e1005659.

(35) Liu, S.; Wu, Y.; Lin, T.; Abel, R.; Redmann, J. P.; Summa, C. M.; Jaber, V. R.; Lim, N. M.; Mobley, D. L. Lead Optimization Mapper: Automating Free Energy Calculations for Lead Optimization. *J. Comput. Aided Mol. Des.* **2013**, *27* (9), 755−770.

(36) Midgley, S.; Bariami, S.; Habgood, M.; Mackey, M. Adaptive Lambda Scheduling: A Method for Computational Efficiency in Free Energy Perturbation Simulations. *J. Chem. Inf. Model.* **2025**, *65* (2), 512−519.

(37) Hedges, L.; Mey, A.; Laughton, C.; Gervasio, F.; Mulholland, A.; Woods, C.; Michel, J. BioSimSpace: An Interoperable Python Framework for Biomolecular Simulation. *J. Open Source Softw.* **2019**, *4* (43), 1831.

(38) Klimovich, P. V.; Shirts, M. R.; Mobley, D. L. Guidelines for the Analysis of Free Energy Calculations. *J. Comput. Aided Mol. Des* **2015**, *29*, 397−411.

(39) Xu, H. Optimal Measurement Network of Pairwise Differences. *J. Chem. Inf. Model.* **2019**, *59* (11), 4720−4728.

(40) Cresset. In *Flare User Manual (Version 10.0.0)*; Cresset Group, 2023.

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